Ultra log concavity of laminar Narayana sums

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AI-assisted proof candidate for mathematical review

Review status. This draft has undergone internal mathematical checks. It has not undergone external peer review, and priority has not been established. No claim is made that the real-rootedness conjecture for stack-sorting fibers has been proved.

Abstract

Let P be a Minkowski sum of coordinate simplices indexed by a laminar family on $[m]$, with nonnegative integral weights and positive singleton weights. Write Q for the total weight. A construction is given that makes the degree-$(Q-m)$ homogenization of

$$x^{-m} \sum_{q \in P \cap \mathbb{Z}^m} \prod_{i=1}^{m} N_{q_i}(x)$$

a Lorentzian polynomial. Consequently, its coefficients are ultra log-concave of order $Q-m$. The proof combines a lattice-point recurrence, a formal Narayana substitution, and the ordinary coefficient-preserving truncation theorem of Ross, Süss, and Wannerer. Through Defant's fertilitope results, it yields log-concavity of the descent distribution on every stack-sorting fiber, the weaker question stated immediately after Conjecture 31 in his 2020 paper. The argument does not establish real-rootedness.

1 Statement and conventions

For $a \geq 1$, use the Narayana polynomial convention

$$N_a(x) = \sum_{j=1}^{a} \frac{1}{a} \binom{a}{j} \binom{a}{j-1} x^j. \tag{1}$$

Thus $N_1(x) = x$, and $N_a(x)/x$ has degree $a-1$, positive coefficients throughout its degree range, and constant and leading coefficients equal to one.

A family $\mathcal{L}$ of nonempty subsets of $[m] = \{1, \ldots, m\}$ is *laminar* if any two members are either disjoint or comparable by inclusion. Assume throughout that $m \geq 1$ and that $\mathcal{L}$ contains every singleton. For a nonempty $I \subseteq [m]$, put

$$\Delta_I = \operatorname{conv}\{e_i : i \in I\}, \qquad P = \sum_{I \in \mathcal{L}} y_I \Delta_I, \qquad Q = \sum_{I \in \mathcal{L}} y_I,$$

where $e_1, \ldots, e_m$ are the coordinate vectors. A zero-weight summand is interpreted as $\{0\}$.

A nonnegative sequence $(a_0, \ldots, a_d)$ is *ultra log-concave of order d* if it has no internal zeros and $(a_j / \binom{d}{j})_{j=0}^{d}$ is log-concave. For $d = 0, 1$, the inequalities are vacuous. This is the degree-sensitive convention used below.

Theorem 1.1. *Suppose $y_I \in \mathbb{Z}_{\geq 0}$ for all $I \in \mathcal{L}$ and $y_{\{i\}} \geq 1$ for all i. Set $d = Q - m$. Then*

$$A_P(x) = x^{-m} \sum_{q \in P \cap \mathbb{Z}^m} \prod_{i=1}^{m} N_{q_i}(x) = \sum_{j=0}^{d} a_j x^j \tag{2}$$

has degree exactly d, all a_j are positive, and the binary form $u^d A_P(v/u)$ is Lorentzian. In particular, $(a_0, \ldots, a_d)$ is ultra log-concave of order d. Its endpoint coefficients satisfy

$$a_0 = a_d = |P \cap \mathbb{Z}^m|.$$

The same conclusion applies if the summand weights are initially nonnegative real numbers, the singleton weights are positive, and P is an integral polytope: Lemma 2.1 shows that all these weights are necessarily integral. Any laminar family can be made a family of intervals by ordering leaves in a depth-first traversal of its containment forest. Thus the interval convention for binary nestohedra in [3, Section 2.6] imposes no additional restriction here after a coordinate permutation.

The stack-sorting consequence is stated and proved in Section 6. In particular, the result concerns a fixed fiber, rather than only a union of fibers or an aggregate descent distribution.

2 Lattice points and unique subtree surpluses

Adjoin $[m]$ to $\mathcal{L}$, with weight zero, if it is absent. This changes neither P nor Q. Regard the resulting family as a rooted tree under inclusion. The children of a nonleaf node I are its maximal proper subnodes. They are pairwise disjoint and cover I, since every singleton is present. Set

$$W_I = \sum_{\substack{J \in \mathcal{L} \\ J \subseteq I}} y_J, \qquad q(I) = \sum_{i \in I} q_i.$$

In particular, $W_{[m]} = Q$.

Lemma 2.1. *If the nonnegative real weights above give an integral polytope P, then every y_I is an integer.*

Proof. For every $S \subseteq [m]$, additivity of the minimum of a linear functional under Minkowski sum gives

$$z(S) := \min_{q \in P} q(S) = \sum_{\substack{I \in \mathcal{L} \\ I \subseteq S}} y_I. \tag{3}$$

Indeed, the minimum of $q(S)$ on Δ_I is one when $I \subseteq S$, and zero otherwise. Since a linear minimum on an integral polytope occurs at an integral vertex, each $z(S)$ is integral. Processing the members of $\mathcal{L}$ in increasing inclusion order and using

$$y_I = z(I) - \sum_{\substack{J \in \mathcal{L} \\ J \subsetneq I}} y_J$$

proves the assertion by induction. In particular, a positive singleton weight is at least one. □

Lemma 2.2 (Subtree feasibility)**.** *A vector $q \in \mathbb{R}^m$ belongs to P if and only if*

$$q([m]) = Q, \qquad q(I) \geq W_I \quad (I \in \mathcal{L}). \tag{4}$$

If q and the weights are integral, the allocations witnessing its membership in the Minkowski sum can be chosen integral.

Proof. Necessity follows from (3), or directly by summing the contributions of all simplices supported inside I.

For sufficiency, regard q_i as capacity available at leaf i. These capacities are nonnegative by the singleton inequalities. Process the nodes in an order in which every proper descendant precedes its ancestors. At node I, allocate a mass of y_I among leaves in I, without exceeding the residual capacity at any leaf. At that time all proper descendants of I have been processed, while any previously processed node not contained in I is disjoint from I. This follows from laminarity and the chosen order. Hence the total previously allocated mass inside I is precisely $W_I - y_I$. The residual capacity there is

$$q(I) - (W_I - y_I) \geq y_I,$$

so the allocation is possible. If capacities and weights are integral, fill the leaves in any fixed order using integral amounts; the residual capacities remain integral.

After processing the root, the total allocated mass is $Q = q([m])$. All residual capacities are nonnegative, so each is zero. The allocation made at I is an element of $y_I\Delta_I$, and the sum of these allocations is q. This proves membership. The construction proves existence; it does not assert uniqueness of a Minkowski decomposition. □

For a node I and $r \in \mathbb{Z}_{\geq 0}$, let $\mathcal{C}_I(r)$ be the integer vectors indexed by I satisfying

$$q(I) = W_I + r, \qquad q(J) \geq W_J \quad \text{for every node } J \subseteq I.$$

These sets are finite because all coordinates are positive and the total is fixed. Define the *reduced* weighted sum

$$F_I(r; x) = \sum_{q \in \mathcal{C}_I(r)} \prod_{i \in I} \frac{N_{q_i}(x)}{x}. \tag{5}$$

Exactly one factor of x is removed for each original leaf. In particular, Lemma 2.2 gives $F_{[m]}(0; x) = A_P(x)$.

Lemma 2.3 (Unique surplus recurrence)**.** *For a singleton of weight a, one has $F_{\{i\}}(r; x) = N_{a+r}(x)/x$. If I has children $I_1, \ldots, I_\ell$ and own weight $c = y_I$, then*

$$F_I(r; x) = \sum_{\substack{r_1, \ldots, r_\ell \geq 0 \\ r_1 + \cdots + r_\ell = c + r}} \prod_{j=1}^{\ell} F_{I_j}(r_j; x). \tag{6}$$

Proof. A feasible parent vector uniquely determines

$$r_j = q(I_j) - W_{I_j} \geq 0.$$

Since the children partition I and $W_I = c + \sum_j W_{I_j}$, their surpluses sum to $c + r$. Its restriction to I_j lies in $\mathcal{C}_{I_j}(r_j)$. Conversely, concatenate one vector from each such child set. All proper-node inequalities hold inside a child, and the parent sum and inequality hold by the surplus identity. These maps are inverse. The weights in (5) multiply across disjoint child blocks, proving (6). No choice of a Minkowski decomposition is counted. □

The zero-weight root also handles a disconnected containment forest: its zero surplus forces all maximal component surpluses to be zero, so their contributions multiply.

3 A finite polynomial model for the recurrence

All generating series in this section are formal power series in z over $\mathbb{Q}[x]$. Put

$$b_r(x) = \frac{N_{r+1}(x)}{x}, \qquad M(z,x) = \sum_{r\geq 0} b_r(x) z^r, \qquad R(u,x) = (1+u)(1+xu).$$

There is a unique series $u = u(z,x) \in z\mathbb{Q}[x][[z]]$ satisfying $u = zR(u,x)$. Lagrange inversion gives, for $r \geq 0$,

$$\begin{aligned}[z^r]\frac{u}{z} &= \frac{1}{r+1}[t^r](1+t)^{r+1}(1+xt)^{r+1} \\ &= \sum_{j=0}^{r} \frac{1}{r+1}\binom{r+1}{j}\binom{r+1}{j+1} x^j = b_r(x).\end{aligned}$$

Consequently

$$u = zM(z,x), \qquad M(z,x) = R(u,x), \qquad z = \frac{u}{R(u,x)}. \tag{7}$$

Equivalently, $M = 1 + (1+x)zM + xz^2M^2$. Because $R(0,x) = 1$, both substitutions in (7) are valid and mutually inverse as formal substitutions. No analytic convergence is required.

For a polynomial $B(u,x)$, define

$$\Theta B = R(u,x)\frac{B(u,x) - B(0,x)}{u}. \tag{8}$$

The quotient is a polynomial. If B has nonnegative coefficients, so does ΘB, because the subtraction removes exactly its terms of u-degree zero. If $G(z,x) = B(u(z,x),x)$, then

$$\frac{G(z,x) - G(0,x)}{z} = (\Theta B)(u(z,x),x). \tag{9}$$

Thus Θ implements one forward shift of the sequence of z-coefficients.

Proposition 3.1. *Define one polynomial $B_I(u,x)$ for each node by*

$$B_{\{i\}} = \Theta^{\,y_{\{i\}}-1}R, \qquad B_I = \Theta^{\,y_I}\left(\prod_{j=1}^{\ell} B_{I_j}\right) \quad \text{at an internal node.} \tag{10}$$

All these polynomials have nonnegative coefficients and satisfy

$$\sum_{r\geq 0} F_I(r;x) z^r = B_I(u(z,x),x).$$

In particular, $A_P(x) = B_{[m]}(0,x)$. The full expression uses m copies of R and exactly $d = Q - m$ applications of Θ.

Proof. At a leaf of weight a, the sequence $N_{a+r}(x)/x = b_{a+r-1}(x)$ is the $(a-1)$-fold forward shift of $(b_r(x))$. Equations (7) and (9) give the first rule. At an internal node, the product of the child generating series has coefficient at z^t equal to the convolution over child surpluses summing to t. Equation (6) retains the coefficients with $t = y_I + r$, which is exactly a y_I-fold shift. This proves the second rule and the generating-series identity by induction. Evaluation at $z = 0$, equivalently $u = 0$, gives the root value. Finally, the shift count is

$$\sum_{i=1}^{m}(y_{\{i\}} - 1) + \sum_{\substack{I\in\mathcal{L}\\ |I|>1}} y_I = Q - m.$$

Nonnegativity follows at each multiplication and each application of (8). □

4 Lorentzian ingredients

We use Lorentzian polynomials with their *ordinary monomial coefficients*, not factorial-normalized coefficients. The following standard facts fix the precise conventions needed here.

For a homogeneous polynomial f of degree $D \geq 2$ with nonnegative coefficients, the characterization in [1, Theorem 2.25] says that f is Lorentzian if and only if its support is M-convex and every derivative $\partial^\alpha f$ with $|\alpha| = D - 2$ has a Hessian with at most one positive eigenvalue. Nonnegative constants and linear forms are Lorentzian; the zero polynomial is allowed in the closure statements. The support condition and a direct check of the truncation used below are detailed in Appendix A.

The required preservation properties are:

(i) products of Lorentzian polynomials are Lorentzian;

(ii) partial derivatives, nonnegative linear substitutions (including setting a variable to zero), and multiplication by a positive scalar preserve the Lorentzian property;

(iii) deleting all monomials whose exponent in one coordinate exceeds a fixed nonnegative integer preserves the Lorentzian property, with the retained coefficients unchanged.

The first two are standard results of Brändén and Huh; see [1, Theorem 2.10 and Corollary 2.32]. The ordinary truncation statement in (iii) is credited to Ross, Süss, and Wannerer, [4, Definition 3.2 and Proposition 3.3]. Their Theorem 3.1 also records the product and substitution closures. Appendix A gives a short verification of the particular upper-coordinate truncation required here; no novelty is claimed for that preservation theorem.

Finally, for $a_j \geq 0$, the binary form

$$\sum_{j=0}^{d} a_j u^{d-j} v^j \tag{11}$$

is Lorentzian if and only if $(a_j/\binom{d}{j})_{j=0}^{d}$ is log-concave with no internal zeros, with the usual conventions in degrees zero and one; see [1, Example 2.26]. Notice that the binomial denominator uses the degree d of this binary form, not the degree of an auxiliary form from which it is extracted.

5 The homogeneous lift and extraction

Consider any subexpression built from R by multiplication and Θ. Let m' denote its number of base copies of R, let d' denote its number of applications of Θ, and let $B(u, x)$ denote its value. Define initially in the Laurent polynomial ring

$$H(u, v) = u^{d'} B(u, v/u). \tag{12}$$

The following simultaneous invariant makes this a polynomial and fixes its exact homogenization degree.

Lemma 5.1. *For every such subexpression, H is a polynomial with nonnegative coefficients. Every monomial of H has total degree at least d'. Its unique highest-total-degree monomial is*

$$(uv)^{m'+d'},$$

with coefficient one. In particular, its total degree is exactly $D' = 2(m' + d')$.

Proof. At a base copy, $m' = 1$, $d' = 0$, and $H = (1+u)(1+v)$, so the assertions hold. Multiplication adds the leaf and shift counts and multiplies the corresponding H polynomials. It therefore preserves all three assertions, including uniqueness and coefficient one of the top monomial.

For a shift, assume the assertions hold for B and write $H_{[d']}$ for the homogeneous degree-d' part of H. A monomial $u^a x^b$ of B becomes $u^{d'+a-b}v^b$ in H and has total degree $d' + a$. Hence

$$H_{[d']}(u, v) = u^{d'} B(0, v/u). \tag{13}$$

For $\widetilde{B} = \Theta B$, direct substitution in (8) gives

$$\widetilde{H}(u, v) := u^{d'+1}\widetilde{B}(u, v/u) = (1 + u)(1 + v)(H - H_{[d']}). \tag{14}$$

Deleting the lowest homogeneous layer leaves a polynomial with nonnegative coefficients and minimum degree at least $d' + 1$. The old top term is not deleted because $2(m' + d') > d'$ for $m' \geq 1$. The unique new top term is therefore $(uv)^{m'+d'+1}$, with coefficient one. This proves the invariant after a shift, and hence for all subexpressions. □

Homogenize at the *actual* degree supplied by the lemma:

$$h(w, u, v) = w^{D'} H(u/w, v/w), \qquad D' = 2(m' + d'). \tag{15}$$

At a base copy, $h = (w + u)(w + v)$. Under multiplication, the degrees D' add, so the homogeneous lifts multiply without an extra power of w.

For an integer $b \geq 0$, let $\mathrm{T}_{w\leq b}$ delete monomials of w-exponent greater than b, keeping all other coefficients unchanged. Since the least degree in H is at least d', the largest possible w-exponent in h is $D' - d'$. Its terms of that exponent correspond exactly to $H_{[d']}$. Equation (14), homogenized in degree $D' + 2$, is therefore the identity

$$\widetilde{h}(w, u, v) = (w + u)(w + v)\ \mathrm{T}_{w\leq D'-d'-1}\, h(w, u, v). \tag{16}$$

Here $D' - d' - 1 = 2m' + d' - 1 \geq 1$. In the componentwise truncation notation of [4, Definition 3.2], this is the cap $(D' - d' - 1, D', D')$ in variable order (w, u, v).

Proposition 5.2. *Every homogeneous lift h in* (15) *is Lorentzian.*

Proof. The base $(w + u)(w + v)$ is a product of nonnegative linear forms. Products preserve the Lorentzian property. A shift is exactly the coefficient-preserving truncation followed by multiplication in (16), both of which preserve it. Induction on the expression proves the claim. □

In particular, this argument does not assume that arbitrary homogenization preserves a Lorentzian property. It constructs each homogeneous lift through the stated closure operations. The exact degree invariant also shows that (16) has no hidden extra power, derivative multiplier, or coefficient normalization.

Proof of Theorem 1.1. Apply the construction to the root expression in Proposition 3.1. Its parameters are $m' = m$, $d' = d = Q - m$, and $D = 2(m + d)$. By (13),

$$\begin{aligned} u^d A_P(v/u) = H_{[d]}(u, v) &= [w^{D-d}]h(w, u, v) \\ &= \frac{1}{(D-d)!}\left.\frac{\partial^{D-d} h}{\partial w^{D-d}}\right|_{w=0}. \end{aligned} \tag{17}$$

The factorial is the ordinary coefficient-extraction factor. Differentiation, specialization at zero, and positive rescaling preserve the Lorentzian property, so the extracted binary form is Lorentzian of degree d.

For completeness, $P \cap \mathbb{Z}^m$ is nonempty: in each integral simplex summand choose an integral vertex and add these vertices. Every lattice point q satisfies $q_i \geq 1$ and $\sum_i q_i = Q$. Thus

$$\prod_i \frac{N_{q_i}(x)}{x}$$

has positive coefficients in every degree from zero through $\sum_i (q_i - 1) = d$, and both endpoint coefficients are one. Summing gives exact degree d, positivity of every coefficient, and $a_0 = a_d = |P \cap \mathbb{Z}^m|$. Applying the binary criterion (11) completes the proof. □

When $d = 0$, all singleton weights are one and every internal weight is zero; $P = \{(1, \ldots, 1)\}$ and $A_P = 1$. Degree one also presents no endpoint problem. Single-leaf trees, zero internal weights, and disconnected families were included throughout.

6 Descent distributions on stack sorting fibers

Let s be West's stack-sorting map. Recursively, $s(\varnothing) = \varnothing$ and $s(LnR) = s(L)s(R)n$, where n is the largest letter. For $\pi \in S_n$, $n \geq 1$, define

$$f_r(\pi) = |\{\sigma \in S_n : s(\sigma) = \pi,\ \mathrm{des}(\sigma) = r\}|.$$

Defant's Conjecture 31 in [2, p. 37] asks for real-rootedness of the corresponding sequence. The paragraph immediately following it explicitly asks the weaker general log-concavity question. The following consequence addresses that weaker question.

Corollary 6.1. *For every $n \geq 1$ and $\pi \in S_n$, the sequence $(f_0(\pi), \ldots, f_{n-1}(\pi))$ is log-concave. More precisely, if $s^{-1}(\pi) \neq \varnothing$ and $k = \mathrm{des}(\pi)$, then $d = n - 2k - 1 \geq 0$, the support of this sequence is exactly*

$$k, k+1, \ldots, n-k-1,$$

and $(f_k(\pi), f_{k+1}(\pi), \ldots, f_{n-k-1}(\pi))$ is ultra log-concave of order d.

Proof. For a nonempty fiber, Defant's fertilitope theorems and refined fertility formula [3, Theorems 3.1–3.2 and the following displayed formula] give an integral binary nestohedron $\mathrm{Fer}_\pi \subseteq \mathbb{R}^{k+1}$, with positive integral singleton weights, such that

$$\sum_{r=0}^{n-1} f_r(\pi) x^{r+1} = \sum_{q \in \mathrm{Fer}_\pi \cap \mathbb{Z}^{k+1}} \prod_{i=1}^{k+1} N_{q_i}(x). \tag{18}$$

Every lattice point has total coordinate sum $n - k$. Consequently, the parameters in Theorem 1.1 are

$$m = k + 1, \qquad Q = n - k, \qquad d = Q - m = n - 2k - 1.$$

After dividing (18) by x^{k+1}, that theorem gives

$$A_{\mathrm{Fer}_\pi}(x) = \sum_{j=0}^{d} f_{k+j}(\pi) x^j$$

with all coefficients positive, and proves the stated ULC conclusion. Positivity and exact degree also give the claimed full support.

To see ordinary log-concavity explicitly, write $a_j = f_{k+j}(\pi)$. For $1 \le j \le d-1$, ULC implies

$$a_j^2 \ge a_{j-1}a_{j+1}\frac{\binom{d}{j}^2}{\binom{d}{j-1}\binom{d}{j+1}} = a_{j-1}a_{j+1}\frac{(j+1)(d-j+1)}{j(d-j)} \ge a_{j-1}a_{j+1}.$$

Padding this positive block by zeros at either end preserves log-concavity. For $d = 0, 1$ the same assertion follows directly. If the fiber is empty, all $f_r(\pi)$ vanish and the log-concavity inequalities hold trivially; no degree-d claim is needed in that case. □

If the empty permutation is included, its unique preimage is itself and its descent distribution is the singleton sequence (1). This is a separate trivial case; the formula $d = n - 2k - 1$ above is asserted only for nonempty permutations with nonempty fibers.

7 Scope and the remaining real rootedness question

The argument proves a ULC statement for the trimmed polynomial. It does not prove the real-rootedness claims in [2, Conjecture 31] or [3, Conjectures 1.1 and 1.2]. The distinction is substantive: ordinary coordinate truncation preserves Lorentzian polynomials but need not preserve real stability.

For example, start from the stable product $(w+u)^2(w+v)^2$ and delete its w^4 term. Specializing $u = v = 1$ gives

$$(w+1)^4 - w^4 = (2w+1)(2w^2+2w+1),$$

whose last factor has nonreal roots $(-1 \pm i)/2$. Thus a closure step used in the lift cannot simply be replaced by a stability argument. This example is not a counterexample to the fiber real-rootedness conjecture: it concerns an auxiliary lift, not a fiber polynomial.

The laminar assumption and positive singleton weights are also essential to the scope of this draft. Overlapping nonlaminar blocks do not have the child-partition recurrence proved here. Zero singleton weights would require conventions for N_0 and a different base analysis. Neither extension is claimed. The mathematical argument is presented for external checking; its publication priority and acceptance remain unestablished.

A A direct check of the truncation step

For clarity, the upper-coordinate case of [4, Proposition 3.3] is verified here using the characterization in [1, Theorem 2.25]. This is a proof of a previously established closure property, not a new truncation theorem.

A nonempty finite set $S \subseteq \mathbb{Z}^t_{\ge 0}$ of constant coordinate sum is M-convex if it satisfies the exchange condition: whenever $\alpha, \beta \in S$ and $\alpha_i > \beta_i$, there is a j with $\alpha_j < \beta_j$ such that $\alpha - e_i + e_j \in S$. This is the standard exchange formulation in the present constant-sum setting.

Let f be homogeneous Lorentzian of degree D, and fix a coordinate z_p and a cap $b \in \mathbb{Z}_{\ge 0}$. Write $g = \mathrm{T}_{z_p \le b} f$. If $g = 0$, there is nothing to prove. To check its support, take $\alpha, \beta \in \mathrm{supp}(g)$ with $\alpha_i > \beta_i$ and choose j by exchange in $\mathrm{supp}(f)$. The exchanged point stays under the cap: if $j \ne p$, its pth coordinate does not increase; if $j = p$, then $\alpha_p < \beta_p \le b$ and hence $\alpha_p + 1 \le b$. Thus $\mathrm{supp}(g)$ is M-convex.

For $D \geq 2$, fix a differential operator ∂^γ with $|\gamma| = D-2$ and put $t = \gamma_p$. Compare the quadratic forms $\partial^\gamma g$ and $\partial^\gamma f$.

- If $t > b$, the former is zero.
- If $t \leq b-2$, the forms are identical: every contributing exponent in the pth coordinate is at most $t+2 \leq b$.
- If $t = b$, the former is the latter with z_p set to zero. Its Hessian is the principal submatrix omitting row and column p, embedded with a zero row and column. Interlacing gives at most one positive eigenvalue.
- If $t = b-1$, only the z_p^2 coefficient is removed. The new Hessian is the old Hessian minus a nonnegative multiple of $e_p e_p^{\mathsf{T}}$. Subtracting a positive semidefinite matrix cannot increase the number of positive eigenvalues.

The Lorentzian characterization therefore applies to g. In degrees zero and one the assertion follows immediately from nonnegativity. This also confirms directly that the operation used in (16) keeps ordinary coefficients and introduces no factorial normalization.

References

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